

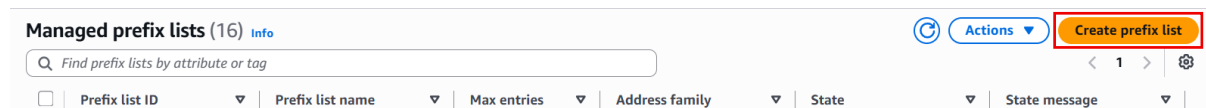
Configuring a Customer Managed Prefix List in AWS

To Begin with the Lab

Summary of the Lab

This lab demonstrates how to configure and use AWS Customer Managed Prefix Lists and AWS Managed Prefix Lists. You create a custom prefix list by defining IPv4 CIDR blocks and reuse it in Security Group inbound rules instead of manually adding multiple IP ranges. The lab also shows how AWS Managed Prefix Lists, such as Amazon CloudFront and Amazon S3, simplify outbound Security Group rules and Route Table configurations by automatically maintaining AWS service IP ranges, improving scalability, consistency, and network management.

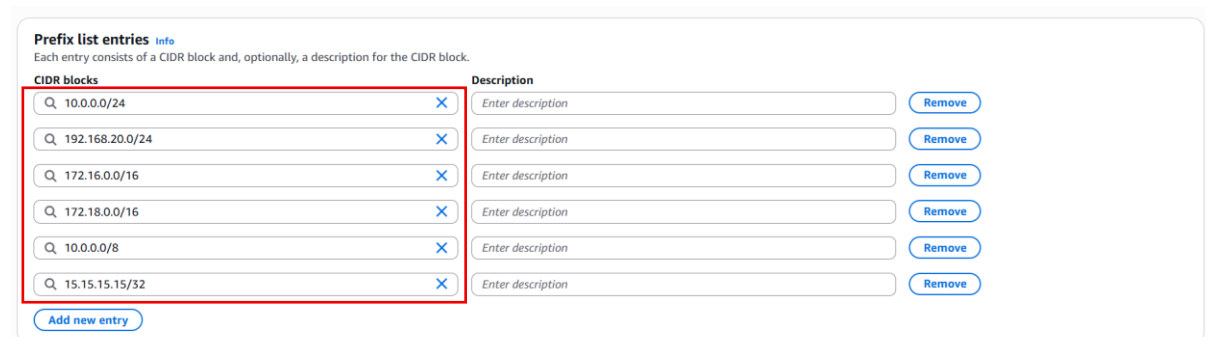
- In the **VPC Console**, select **Managed Prefix Lists** from the left navigation pane.
- Click **Create Prefix List** to create your own Customer Managed Prefix List.



- Provide a meaningful name for the Prefix List.
- Specify the **Maximum Entries**.
- Choose **IPv4** if you are storing IPv4 CIDR blocks.



- Add the required CIDR blocks to the Prefix List.



- Review the Prefix List configuration.
- Click **Create Prefix List** to save the configuration.
- Navigate to the **Security Groups** section in the VPC console.

- Edit the **Inbound Rules** of the Security Group.

sg-06a6a32d414aa8b40 - launch-wizard-6 Actions

Details

Security group name
launch-wizard-6

Owner
463646775279

Security group ID
sg-06a6a32d414aa8b40

Inbound rules count
2 Permission entries

Description
launch-wizard-6 created 2026-07-03T09:42:16.822Z

Outbound rules count
1 Permission entry

VPC ID
vpc-099792461c7d0a616

Inbound rules | Outbound rules | Sharing | VPC associations | Related resources - new | Tags

Inbound rules (2) Manage tags Edit inbound rules

Search

☐	Name	Security group rule ID	IP version	Type	Protocol	Port range
☐	-	sgr-0103d423750c16bee	IPv4	SSH	TCP	22
☐	-	sgr-0bc44c9e4df5f2160	IPv4	HTTP	TCP	80

- Instead of manually entering multiple CIDR blocks, add the required port, such as **HTTP (Port 80)**.
- For the **Source**, select **Prefix List**.

Edit inbound rules Info

Inbound rules control the incoming traffic that's allowed to reach the instance.

Inbound rules Info

Security group rule ID

sgr-0103d423750c16bee

Type

SSH

Protocol

TCP

Port range

22

Source

Custom

0.0.0.0/0

Description - optional

Add rule Delete

Rules with source of 0.0.0.0/0 or ::/0 allow all IP addresses to access your instance. We recommend setting security group rules to allow only the necessary IP addresses.

Prefix lists

- cf-dc-ip-list | pl-0cda641e12d0d7810
- com.amazonaws.us-east-1.dynamodb | pl-02cd2c6b
- com.amazonaws.global | pl-02cd2c6b

Cancel Preview changes Save rules

Understanding AWS Managed Prefix Lists

- Navigate to the **Security Groups** section of the VPC console.
- Create a new Security Group or open an existing Security Group whose outbound traffic you want to control.
- Edit the **Outbound Rules** of the Security Group.

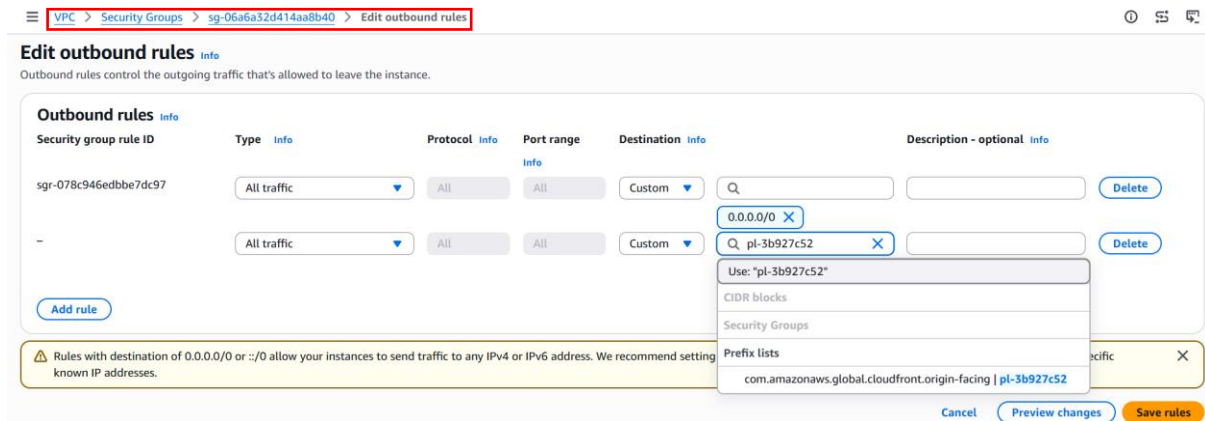
sg-0032f8736b608a77d - launch-wizard-7 Details Inbound rules Outbound rules Sharing VPC associations Related resources - new Tags

Outbound rules (1) Manage tags Edit outbound rules

Search

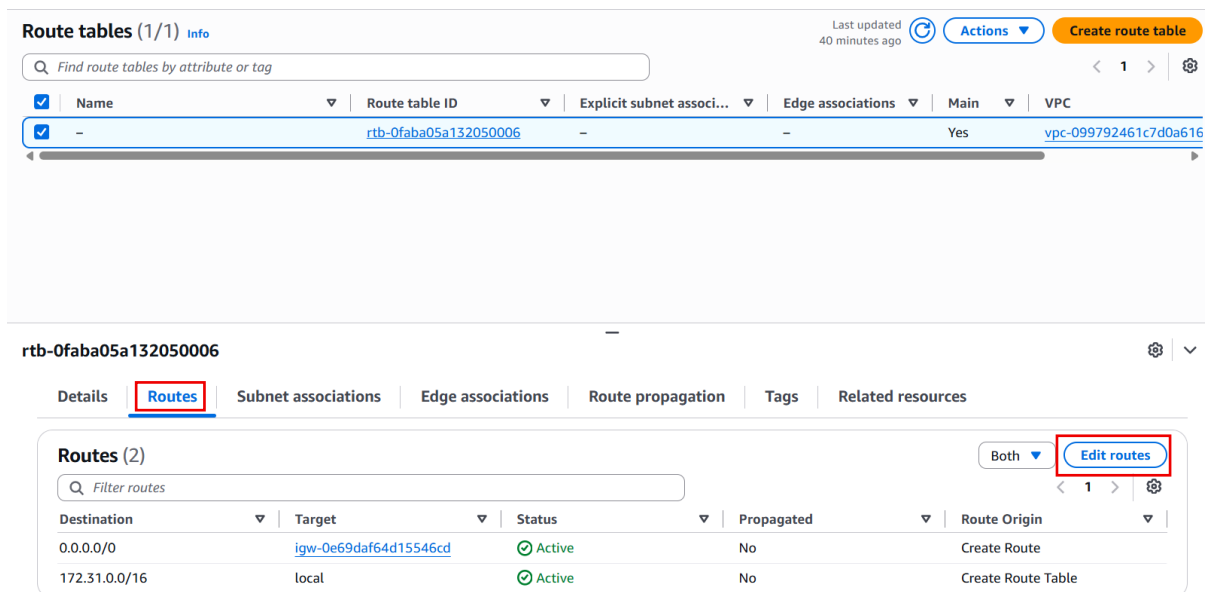
☐	Name	Security group rule ID	IP version	Type	Protocol	Port range
☐	-	sgr-0b50f38714e055e87	IPv4	All traffic	All	All

- Instead of allowing traffic to a CIDR block, choose **Prefix List** as the destination.
- choose the **Amazon CloudFront** Prefix List.
- Save the Security Group rule.
- This allows the EC2 instance to communicate only with Amazon CloudFront without requiring you to know its IP address ranges.



Use an AWS Managed Prefix List in a Route Table

- Navigate to the **Route Tables** section in the VPC console.
- Select the Route Table that should forward traffic to a specific AWS service.
- Edit the Route Table.



- When adding a new route, select an **AWS Managed Prefix List** instead of manually entering a destination CIDR block.
- Choose the **Amazon S3 Prefix List** if you want traffic destined for Amazon S3 to follow a specific route.
- Save the Route Table.

Edit routes

Destination	Target	Status	Propagated	Route Origin
172.31.0.0/16	local	Active	No	CreateRouteTable
Q 0.0.0.0/0	Internet Gateway	Active	No	CreateRoute
Q pl-63a5400a	igw-0e69daf64d15546cd	-	No	CreateRoute
instance-connect	Instance	-	No	CreateRoute

pl-0e4bcff02b13bef1e (com.amazonaws.us-east-1.ec2-instance-connect)

pl-0e5696d987d033653 (com.amazonaws.global.groundstation)

pl-2c698745 (com.amazonaws.us-east-1.ipv6.s3express)

pl-3b927c52 (com.amazonaws.global.cloudfront.origin-facing)

pl-480ee021 (com.amazonaws.us-east-1.ipv6.dynamodb)

pl-63a5400a (com.amazonaws.us-east-1.s3)

pl-d928c6b0 (com.amazonaws.us-east-1.s3express)

pl-da1df3b3 (com.amazonaws.us-east-1.ipv6.s3)

Cancel

Preview

Save changes